

## MAKURAZAKI, Japan

WMO: 478310

Lat: 31.2719N

Lon: 130.2931E

Elev: 31

StdP: 100.95

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 1.4        | 2.6 | -5.3                          | 2.4 | 5.4  | -4.0 | 2.7 | 5.7  | 11.5                  | 10.0 | 10.7 | 9.2  | 4.1                   | 0    | 0.444 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.8                    | 32.1            | 25.9 | 31.3 | 25.7 | 30.6 | 25.6 | 27.2                | 30.1 | 26.7 | 29.6 | 26.3 | 29.2 | 4.8                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.5                            | 22.1 | 29.0 | 26.0 | 21.4 | 28.7 | 25.5 | 20.8 | 28.4 | 86.3          | 30.1 | 84.3 | 29.6 | 82.4 | 29.3 | 29.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.8              | 9.4  | 8.4 | -0.7                       | 33.7 | 0.9                | 0.8 | -1.3                                               | 34.3 | -1.9       | 34.7 | -2.4       | 35.2 | -3.0       | 35.8 |    |    |
|                   |      |     | -2.3                       | 27.6 | 0.8                | 0.8 | -2.9                                               | 28.2 | -3.4       | 28.6 | -3.8       | 29.1 | -4.5       | 29.7 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 18.3  | 8.7   | 9.8   | 12.4  | 16.3  | 20.1  | 23.3  | 27.2  | 28.0  | 25.6  | 21.1  | 15.8  | 10.9 | (6)  |
| (7)  |                                                                   | DBStd       | 7.13  | 3.05  | 3.42  | 3.14  | 2.75  | 1.95  | 2.00  | 1.52  | 1.24  | 2.06  | 2.72  | 3.34  | 3.34 | (7)  |
| (8)  |                                                                   | HDD10.0     | 144   | 61    | 42    | 12    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 28   | (8)  |
| (9)  |                                                                   | HDD18.3     | 1124  | 298   | 239   | 184   | 71    | 6     | 0     | 0     | 0     | 0     | 7     | 88    | 231  | (9)  |
| (10) |                                                                   | CDD10.0     | 3183  | 22    | 36    | 87    | 191   | 314   | 398   | 534   | 467   | 343   | 176   | 56    |      | (10) |
| (11) |                                                                   | CDD18.3     | 1122  | 0     | 0     | 1     | 11    | 62    | 148   | 276   | 301   | 217   | 92    | 14    | 1    | (11) |
| (12) |                                                                   | CDH23.3     | 9547  | 0     | 0     | 0     | 8     | 148   | 754   | 2870  | 3426  | 1886  | 434   | 22    | 0    | (12) |
| (13) |                                                                   | CDH26.7     | 2748  | 0     | 0     | 0     | 0     | 5     | 67    | 871   | 1268  | 496   | 42    | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 4.1   | 4.6   | 4.7   | 4.7   | 4.4   | 3.9   | 3.6   | 3.7   | 4.0   | 3.9   | 3.9   | 4.1   | 4.5  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 2353  | 95    | 114   | 166   | 193   | 202   | 536   | 300   | 203   | 220   | 102   | 120   | 101  | (15) |
| (16) |                                                                   | PrecMax     | 3802  | 194   | 270   | 303   | 467   | 358   | 1177  | 996   | 551   | 589   | 377   | 263   | 179  | (16) |
| (17) |                                                                   | PrecMin     | 1556  | 28    | 35    | 49    | 50    | 48    | 199   | 24    | 42    | 28    | 16    | 47    | 9    | (17) |
| (18) |                                                                   | PrecStd     | 527   | 42    | 61    | 61    | 103   | 87    | 201   | 204   | 126   | 145   | 84    | 56    | 47   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 19.6  | 20.4  | 21.6  | 24.5  | 27.4  | 29.4  | 32.9  | 33.4  | 31.7  | 29.2  | 25.2  | 21.5 | (19) |
| (20) |                                                                   | 0.4%        | MCWB  | 16.5  | 17.4  | 17.0  | 18.7  | 19.8  | 25.4  | 25.8  | 26.1  | 25.1  | 23.9  | 21.3  | 17.6 | (20) |
| (21) |                                                                   | 2%          | DB    | 17.6  | 19.1  | 20.2  | 22.9  | 25.9  | 28.4  | 31.8  | 32.4  | 30.7  | 27.8  | 23.9  | 19.7 | (21) |
| (22) |                                                                   | 2%          | MCWB  | 14.7  | 16.4  | 16.4  | 18.1  | 19.6  | 24.5  | 25.8  | 26.0  | 24.9  | 23.0  | 19.3  | 16.2 | (22) |
| (23) |                                                                   | 5%          | DB    | 16.0  | 17.8  | 19.3  | 22.0  | 24.9  | 27.4  | 31.0  | 31.6  | 30.1  | 26.8  | 22.8  | 18.3 | (23) |
| (24) |                                                                   | 5%          | MCWB  | 12.8  | 14.9  | 15.5  | 17.7  | 19.6  | 24.0  | 25.8  | 25.9  | 24.6  | 21.9  | 18.6  | 14.7 | (24) |
| (25) |                                                                   | 10%         | DB    | 14.2  | 16.1  | 18.2  | 21.1  | 24.0  | 26.6  | 30.2  | 31.0  | 29.2  | 25.8  | 21.6  | 16.8 | (25) |
| (26) |                                                                   | 10%         | MCWB  | 10.8  | 12.8  | 14.6  | 17.1  | 19.4  | 23.6  | 25.7  | 25.8  | 24.2  | 20.9  | 17.6  | 13.3 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 17.3  | 18.4  | 19.0  | 21.6  | 23.7  | 26.5  | 27.7  | 28.1  | 27.2  | 25.6  | 22.6  | 18.8 | (27) |
| (28) |                                                                   | 0.4%        | MCDWB | 19.0  | 19.8  | 20.0  | 23.2  | 25.4  | 28.3  | 30.7  | 31.0  | 29.7  | 27.9  | 24.4  | 20.2 | (28) |
| (29) |                                                                   | 2%          | WB    | 15.2  | 16.8  | 17.6  | 20.3  | 22.5  | 25.7  | 27.1  | 27.4  | 26.5  | 24.2  | 20.7  | 16.9 | (29) |
| (30) |                                                                   | 2%          | MCDWB | 17.4  | 18.7  | 19.5  | 21.6  | 24.0  | 27.3  | 29.9  | 30.3  | 29.1  | 26.8  | 22.9  | 19.1 | (30) |
| (31) |                                                                   | 5%          | WB    | 13.4  | 15.2  | 16.4  | 19.1  | 21.6  | 25.2  | 26.7  | 26.9  | 25.8  | 23.2  | 19.5  | 15.3 | (31) |
| (32) |                                                                   | 5%          | MCDWB | 15.5  | 17.5  | 18.6  | 20.8  | 23.3  | 26.7  | 29.4  | 29.9  | 28.6  | 26.0  | 21.9  | 17.7 | (32) |
| (33) |                                                                   | 10%         | WB    | 11.2  | 13.2  | 15.1  | 18.1  | 20.7  | 24.6  | 26.2  | 26.5  | 25.1  | 21.9  | 18.3  | 13.7 | (33) |
| (34) |                                                                   | 10%         | MCDWB | 13.9  | 15.6  | 17.7  | 20.3  | 22.8  | 26.1  | 28.9  | 29.6  | 28.2  | 24.7  | 20.9  | 16.3 | (34) |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 7.1   | 7.5   | 7.8   | 7.6   | 7.2   | 5.3   | 5.3   | 5.8   | 6.6   | 7.6   | 7.8   | 7.6  | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 9.3   | 8.7   | 8.6   | 8.2   | 8.5   | 5.9   | 6.5   | 6.8   | 7.0   | 7.8   | 8.1   | 8.8  | (36) |
| (37) |                                                                   | 5% DB       | MCWBR | 7.4   | 7.3   | 6.3   | 5.1   | 4.5   | 3.0   | 2.7   | 2.7   | 3.0   | 4.0   | 5.1   | 6.4  | (37) |
| (38) |                                                                   | 5% WB       | MCDBR | 8.1   | 7.5   | 7.1   | 5.6   | 5.1   | 4.1   | 5.4   | 5.7   | 5.6   | 6.0   | 6.3   | 7.5  | (38) |
| (39) |                                                                   | 5% WB       | MCWBR | 7.4   | 7.2   | 6.7   | 5.1   | 4.2   | 2.8   | 2.6   | 2.7   | 2.9   | 3.8   | 5.0   | 6.4  | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.440 | 0.475 | 0.529 | 0.544 | 0.547 | 0.546 | 0.520 | 0.512 | 0.500 | 0.473 | 0.439 | 0.431 | (40) |      |
| (41) |                                                                   | taud        | 2.081 | 2.017 | 1.887 | 1.905 | 1.964 | 2.060 | 2.165 | 2.194 | 2.177 | 2.161 | 2.176 | 2.106 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 757   | 771   | 759   | 766   | 766   | 764   | 782   | 785   | 777   | 766   | 755   | 741   | (42) |      |
| (43) |                                                                   | Edh at Noon | 135   | 157   | 192   | 195   | 185   | 168   | 151   | 145   | 142   | 135   | 122   | 125   | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 2.23  | 2.90  | 3.96  | 4.71  | 4.92  | 4.11  | 5.54  | 5.55  | 4.56  | 3.74  | 2.69  | 2.14  | (44) |      |
| (45) |                                                                   | RadStd      | 0.22  | 0.30  | 0.28  | 0.46  | 0.55  | 0.63  | 0.60  | 0.37  | 0.47  | 0.42  | 0.27  | 0.22  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (16 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page